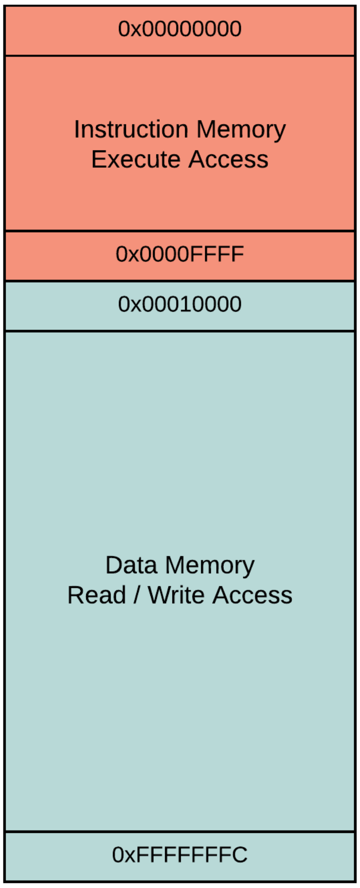
VisUAL
A highly visual ARM emulator

Memory Map

The following diagram shows the general configuration of the VisUAL memory map:



The exact distribution of memory between resources is hardware-specific. For example, in the ARM Cortex M3, the lower 512MB of memory addresses are reserved as [code memory](#).

Since VisUAL does not emulate external device configuration, the memory model for VisUAL is a lot simpler: the lower memory addresses are reserved for saving instructions. As a result the program counter starts with instruction 1 at address `0x0`.

During execution, this instruction space memory is not available for read or write access. The addresses can still be used to manually invoke a branch, such as returning from a subroutine using the instruction `MOV PC, LR`.

Instruction Memory Configuration

By default, VisUAL sets the instruction memory size automatically to fit the available code. This scaling is done in blocks of `256 bytes`. This mode enables the possibility of using the `ADR` psuedo-instruction to load symbol addresses to registers.

The instruction memory size can also be specified manually. The default manual specification is `0x10000 bytes`, which allows emulation of up to 16,384 lines of code.

Memory Access Modes

VisUAL supports two memory access modes, **Open** and **Strict**. In the open access mode, all data memory addresses have read/write access. This mode is enabled by default.

In strict access mode, only data memory addresses explicity defined using the `DCD`, `DCB` or `FILL` directives have write access. All other addresses only have read access.

These modes can be configured via the settings panel.